

Evaluating the Effectiveness of a Behaviour-Aware Multilingual Voice-Based Local Service Discovery System Using a Multi-Agent AI Framework

RESEARCH GROUP ID : R26 -IT- 109

PROGRESS PRESENTATION 1

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Smart Accessible Learnina Environment



Problem

- Existing local service discovery platforms mainly depend on typed English interactions, creating accessibility barriers for many Sri Lankan users.
- Users often communicate using Sinhala-English mixed language and incomplete emergency phrases, which traditional systems fail to understand.
- Non-technical service providers struggle with complex digital onboarding, registration, and password-based authentication processes.
- Current platforms lack intelligent urgency detection, fair workforce distribution, and real-time adaptive provider coordination.

Impact

- Delays in finding urgent local services during emergency situations.
- Reduced accessibility for rural, elderly, and non-technical users.
- Unequal job distribution among service providers, causing income instability.
- Increased frustration due to poor communication accuracy and unreliable provider matching.
- Limited digital inclusion and employment sustainability within local service ecosystems.

Design Excellence

- Behaviour-aware multilingual voice-based service discovery architecture.
- Multi-agent AI framework for intelligent provider coordination and decision-making.
- Hybrid intelligence using speech recognition, NLP, machine learning, geospatial intelligence, and voice biometrics.
- Real-time urgency detection from acoustic speech features and fragmented voice inputs.
- Fairness-aware workforce intelligence framework for equitable job allocation and reputation evolution.
- Zero-barrier voice-driven onboarding and authentication system designed for non-technical workers.
- Personalized AI-driven employee dashboard generation using voice data and selfie-based identity verification.



Adaptive Multilingual and Urgency-Aware Voice Intent Model

SPECIALIZATION:INFORMATION TECHNOLOGY

IT 22 2743 66 | IHALAGEDARA H.T.B

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Adaptive Multilingual and Urgency-Aware Voice Intent Model



How the Solution Addresses the Sub-Problem

- Addresses limitations of existing voice assistants in understanding Sri Lankan multilingual speech patterns.
- Supports Sinhala-English and mixed language communication (code-switching).
- Handles incomplete and fragmented emergency speech inputs such as “Hospital ekak… urgent…”.
- Uses AI-based speech recognition and Natural Language Processing (NLP) for accurate intent understanding.
- Extracts acoustic features such as pitch, speech rate, energy levels, and MFCC features to detect urgency levels.
- Classifies urgency into Normal, Moderate, and High priority categories for faster service response.
- Optional multimodal image input improves service classification accuracy for complex problems.
- Enables real-time intelligent service request understanding in local Sri Lankan contexts.

User Requirements Addressed by the Solution

- Enables natural multilingual voice interaction without requiring typed input.
- Supports Sinhala-English and Tamil-English mixed speech communication.
- Detects urgent situations and prioritizes emergency service requests automatically.
- Understands incomplete or interrupted speech during stressful situations.
- Reduces complexity for non-technical and rural users.
- Improves speed and accuracy of local service discovery.
- Provides more personalized and context-aware service recommendations.
- Enhances accessibility and usability for users with limited digital literacy.

Adaptive Multilingual and Urgency-Aware Voice Intent Model



Design Excellence / Contribution

- Behaviour-aware multilingual voice intelligence model designed for Sri Lankan local speech patterns.
- Hybrid AI approach combining speech recognition, NLP, acoustic analysis, and urgency classification.
- Supports Sinhala-English and Tamil-English code-switching for real-world conversational interaction.
- Intelligent fragmented speech handling for emergency situations and incomplete voice commands.
- Acoustic feature extraction using pitch, speech rate, energy levels, and MFCC features.
- Real-time urgency prediction model classifying requests into Normal, Moderate, and High urgency levels.
- Optional multimodal enhancement using image-based context refinement for better service classification accuracy.
- Modular and scalable AI architecture that can integrate with future smart service ecosystems.

User Feedback on Prototype

- Users found multilingual voice interaction natural and easy to use.
- System successfully recognized Sinhala-English mixed speech commands.
- Emergency and urgent requests were identified with good accuracy.
- Users appreciated the ability to speak naturally without formal sentence structures.
- Fragmented speech handling improved usability during stressful situations.
- Voice-based interaction reduced complexity for non-technical users.
- Users suggested improving support for additional regional accents and dialects.
- Recommended expanding the dataset for better recognition accuracy in noisy environments.



Adaptive Multi-Agent Coordination Framework

SPECIALIZATION: INFORMATION TECHNOLOGY

IT 22 1757 48 | NIMSARA K.A.S

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Adaptive Multi-Agent Coordination Framework



How the Solution Addresses the Sub-Problem

- Addresses limitations of traditional service discovery systems that rely on static provider matching.
- Uses a cooperative multi-agent AI framework for intelligent provider coordination and decision-making.
- Context Agent analyzes user location, request time, and urgency level.
- Search Agent retrieves suitable service providers based on service category and nearby availability.
- Distance Agent calculates real-time travel distance and estimated arrival time.
- Availability Agent verifies whether providers are online, free, and capable of handling urgent tasks.
- Ranking Agent evaluates providers using trust score, ratings, reliability, response speed, and urgent-task performance.
- Enables faster, smarter, and more reliable local service provider selection in real time.

User Requirements Addressed by the Solution

- Provides fast and accurate service provider recommendations.
- Supports urgency-aware provider prioritization during emergency situations.
- Reduces waiting time by selecting nearby and available providers.
- Improves service reliability using AI-based ranking and trust evaluation.
- Ensures better user experience through intelligent real-time coordination.
- Supports dynamic provider reassignment if a selected provider becomes unavailable.
- Reduces manual searching and decision-making for users.
- Enhances transparency and efficiency in local service discovery systems.

Adaptive Multi-Agent Coordination Framework



Design Excellence / Contribution

- Intelligent multi-agent AI architecture for adaptive service provider coordination.
- Hybrid decision-making approach combining contextual analysis, geospatial intelligence, availability checking, and AI-based ranking.
- Context-aware provider selection using location, urgency level, and request timing.
- Real-time travel distance and estimated arrival time calculation for efficient provider matching.
- Dynamic provider ranking model based on trust score, ratings, reliability, and urgent-task performance.
- Adaptive provider reassignment mechanism improves service continuity when providers become unavailable.
- Modular and scalable multi-agent framework suitable for large-scale local service ecosystems.
- Improves overall efficiency, transparency, and reliability of local service discovery systems.

User Feedback on Prototype

- Users experienced faster and more accurate service provider recommendations.
- Nearby provider selection significantly reduced waiting time.
- Urgent service requests were prioritized effectively.
- Users found the intelligent provider ranking system reliable and trustworthy.
- Real-time availability checking improved overall service efficiency.
- Dynamic reassignment feature helped avoid service delays when providers were unavailable.
- Users appreciated the simplicity and responsiveness of the system.
- Suggested improvement: integrate live traffic data and predictive ETA calculations for higher accuracy.





AI-Driven Employment Empowerment & Workforce Intelligence Framework

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IT 22 2455 88 | GAMAGE D.G.P.U

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AI-Driven Employment Empowerment & Workforce Intelligence Framework



How the Solution Addresses the Sub-Problem

- Addresses unfair job allocation and workforce imbalance in existing service platforms.
- Uses AI-driven workforce intelligence to monitor provider activity, workload, and performance.
- Fairness-aware dynamic task allocation balances trust score, distance, availability, and opportunity distribution.
- Dynamic reputation evolution model updates provider trust scores based on recent service quality and urgent-task reliability.
- Machine learning models predict future workforce demand and identify high-demand service areas.
- Income stability prediction helps providers improve long-term earning opportunities and financial sustainability.
- Employee dashboard maintains verified provider profiles, service history, ratings, and workforce analytics.
- Enables sustainable and intelligent workforce coordination within local service ecosystems.

User Requirements Addressed by the Solution

- Provides fair and balanced service opportunities for workers.
- Reduces job concentration among a small group of providers.
- Improves transparency and trust through AI-based reputation scoring.
- Helps workers identify demand hotspots and skill development opportunities.
- Supports long-term income stability and employment sustainability.
- Enables providers to track performance, ratings, and service history through dashboards.
- Improves overall service reliability and workforce efficiency.
- Enhances digital empowerment and accessibility for local gig workers.

AI-Driven Employment Empowerment & Workforce Intelligence Framework



Design Excellence / Contribution

- AI-driven workforce intelligence framework designed for fair and sustainable gig-economy management.
- Hybrid optimization approach combining fairness-aware task allocation, reputation analysis, and demand prediction.
- Dynamic reputation evolution model adapts provider trust scores using recent performance and urgent-task reliability.
- Machine learning-based workforce demand prediction identifies future service demand hotspots.
- Income stability prediction model provides recommendations to improve long-term provider earnings.
- Employee dashboard integrates provider profiles, service analytics, ratings, and workforce insights.
- Real-time workforce monitoring improves transparency, efficiency, and service reliability.
- Modular and scalable framework suitable for large-scale local service ecosystems.

User Feedback on Prototype

- Providers appreciated the fairer distribution of service opportunities.
- Users found the AI-based ranking and trust scoring system reliable and transparent.
- Workforce dashboard helped providers monitor performance and service history easily.
- Demand prediction insights were useful for identifying high-demand service areas.
- Providers felt the system supported long-term income stability and employment sustainability.
- Real-time analytics improved overall workforce coordination and efficiency.
- Users found the interface simple and easy to understand.
- Suggested improvement: include more advanced financial analytics and personalized skill recommendations.



Automated Profile Generation and Zero-Barrier Onboarding Framework for Non-Technical Service Providers

SPECIALIZATION: INFORMATION TECHNOLOGY

IT 22 2665 14 | UDAYANGANI K.M.I

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Automated Profile Generation and Zero-Barrier Onboarding Framework for Non-Technical Service Providers

How the Solution Addresses the Sub-Problem

- Addresses digital accessibility barriers faced by non-technical service providers in Sri Lanka.
- Eliminates the need for typing usernames, passwords, and email addresses during onboarding.
- Uses an AI-based multilingual voice registration system supporting Sinhala, and English.
- Speech-to-text technology automatically converts spoken responses into structured profile data.
- Voice biometric authentication enables secure password-free login.
- OTP-based mobile verification provides additional identity security and fallback authentication.
- Partial registration mechanism allows providers to pause and continue registration later.
- Duplicate identity detection prevents account conflicts in shared-device environments.
- Selfie-based profile generation automatically creates personalized employee dashboards.
- Enables fully voice-driven onboarding and authentication for users with limited digital literacy.

User Requirements Addressed by the Solution

- Enables simple and natural voice-based registration without typing.
- Supports multilingual voice interaction using Sinhala, Tamil, and English.
- Reduces complexity for non-technical and rural service providers.
- Provides secure password-free authentication using voice biometrics and OTP verification.
- Allows users to continue incomplete registration sessions later.
- Automatically generates professional employee profiles and dashboards.
- Improves accessibility for workers with limited digital experience.
- Enhances trust, security, and usability within the service platform ecosystem.

Automated Profile Generation and Zero-Barrier Onboarding Framework for Non-Technical Service Providers



Design Excellence / Contribution

- Voice-driven zero-barrier onboarding framework designed specifically for non-technical service providers.
- Hybrid authentication approach combining speech recognition, voice biometrics, OTP verification, and selfie-based identity validation.
- Multilingual conversational voice interface supporting Sinhala, Tamil, and English interactions.
- Password-free authentication system reduces complexity and improves accessibility.
- Adaptive partial registration mechanism allows users to pause and resume onboarding sessions.
- Duplicate identity detection and conflict resolution improve account integrity in shared-device environments.
- Selfie-based automatic dashboard generation creates professional employee profiles with zero manual configuration.
- Modular and scalable architecture suitable for large-scale local service ecosystems and low digital-literacy communities.

User Feedback on Prototype

- Users found the voice-guided registration process simple and easy to understand.
- Non-technical service providers successfully completed onboarding without typing assistance.
- Multilingual voice prompts improved user comfort and accessibility.
- Voice biometric login was considered fast and convenient compared to password-based systems.
- OTP fallback authentication improved user trust and system reliability.
- Automatic profile and dashboard generation reduced setup complexity for users.
- Users appreciated the ability to pause and continue registration later.
- Suggested improvement: improve voice recognition accuracy in noisy outdoor environments and support additional regional accents.





THANK YOU !